



MMHA: A multi-modal data fusion algorithm with hierarchical attention for disease diagnosis and cancer subtype prediction

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ABSTRACT

In precision medicine, multiomics data integration provides a comprehensive understanding of biological processes, making it essential for accurate disease diagnosis and biomarker identification. However, most existing approaches rely on graph neural networks to combine and propagate information, often integrating additional models to create multi-modal representations of patients. While effective in capturing common associations between individuals, these graph-based methods often overlook critical differences among disease subtypes. To address this limitation, we propose a novel end-to-end interpretable method for multiomics integration, designed to enhance cancer subtype prediction. Our approach employs a Multi-Modal Hierarchical Attention (MMHA) mechanism to handle the complexity of multi-source data with varying scales and structures. Unlike graph-based methods, our model uses the adaptive weighting capabilities of hierarchical attention to aggregate features from different data modalities, uncovering relationships and complementarities without the need for predefined graph structures. Extensive experiments on six public datasets spanning cancer and neurodegenerative disease subtypes demonstrate that our method significantly outperforms state-of-the-art techniques. For instance, MMHA improves accuracy by 4.5 % on the LGG dataset and achieves near-perfect performance with 99.09 % accuracy on the KIPAN dataset, highlighting its robustness in diverse scenarios. Moreover, it effectively identifies key biomarkers, offering valuable insights to support the development of personalized treatment plans and advancing precision medicine.

1. Introduction

The surge in multi-modal biomedical data has significantly advanced research in computer-aided diagnosis in recent years. Unlike single-modal data, heterogeneous data from various modalities provides complementary perspectives on a patient's condition, enabling more accurate and reliable diagnoses (Boehm, Khosravi, Vanguri, Gao, & Shah, 2022; Lan et al., 2024; Zhou et al., 2023). However, effectively integrating and utilizing such data is crucial not only for improving diagnostic precision but also for addressing the challenges posed by data heterogeneity and complexity. Developing efficient multi-modal data integration methods has thus become a central focus in precision medicine research.

Building upon recent successes in multi-modal deep learning, the integration of diverse clinical, imaging, and molecular data has been increasingly applied across various cancer types, showing great potential in advancing clinical precision medicine. By combining clinical,

imaging, and molecular data, researchers have successfully developed comprehensive multi-omics models for predicting survival outcomes in pancreatic cancer (Osipov et al., 2024). In breast cancer, deep learning systems integrating mammography, ultrasound imaging, and clinical metadata have improved the accuracy of risk stratification and clinical management (Qian et al., 2025). For HER2-positive gastric cancer, multimodal frameworks have been established to effectively predict patient responses to anti-HER2 monotherapy and combination immunotherapy (Zheng et al., 2025). In the study of infectious diseases, multimodal integration approaches have enabled rapid pathogen identification and accurate prognosis prediction for pulmonary infections (Vanguri et al., 2022). Collectively, these advances have significantly promoted the clinical adoption of multimodal learning in precision medicine.

Early approaches to multi-modal deep learning have primarily focused on cancer subtype diagnosis and prognosis prediction, employing techniques such as ensemble learning, non-negative matrix factorization, and multi-kernel learning. Methods such as ensemble learning

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(Barnes et al., 2023; Ying, Yang, & Zhou, 2023), non-negative matrix factorization (Lan et al., 2023; Peng et al., 2020), and multi-kernel learning (Liu, Zhou, Shen, & Yin, 2013; Wen et al., 2017; Zhang et al., 2011) have been widely adopted for effectively integrating diverse data types. For instance, Arya and Saha (2022) proposed a stacked ensemble framework combining gene expression, copy number variations, and clinical data to enhance breast cancer prognosis prediction. Similarly, Du and Zhao (2023) introduced a multi-modal adversarial representation learning framework to reduce modality gaps and improve prediction accuracy. However, most of these methods focus primarily on feature representation and patient similarity, while insufficiently leveraging complementary information across modalities. To address this limitation, graph-based models, particularly Graph Convolutional Networks (GCNs), have emerged as promising tools for integrating multi-modal data and analyzing patient relationships. GCNs represent patient similarities through graphs and aggregate information from connected nodes, making them effective in various biomedical applications, including Alzheimer's disease prediction (Marinescu et al., 2019; Zhu, Du, & Yan, 2019), autism diagnosis (Cao et al., 2021; Craddock et al., 2013), and cancer prognosis prediction (Gao et al., 2020).

Despite their success, GNN-based methods (Li et al., 2022; Wang et al., 2021; Yin et al., 2022) often rely on latent multi-modal representations to construct graphs, overlooking other valuable features. These representations are typically optimized to minimize empirical loss, but such optimization can distort the underlying patient graph due to non-linear mappings. Moreover, constructing graphs based on latent features often results in overly dense connections, leading to over-smoothing during information aggregation (Chen, Wei, Huang, Ding, & Li, 2020). This over-smoothing reduces the discriminative power of node features and degrades classification performance, as nodes from the same class become overly similar. Although graph neural networks (GNNs) have been widely adopted for modeling relationships in biomedical data, they often face limitations such as limited scalability, over-smoothing, and difficulties in handling heterogeneous modalities.

In recent studies, increasing attention has been given to non-graph-based approaches for multi-modal data integration. MOINER (Zhang et al., 2024) enhances selected features through sample similarity networks, projects them into two-dimensional representations based on feature similarity and the J-V algorithm, and employs an ensemble Vision Transformer for classification and biomarker identification. Similarly, MoSwformer (Liu, Zhu, Li, Shi, & Wang, 2024) combines sliding window Transformer embeddings with modality-specific attention mechanisms and representation learning modules to strengthen multi-omics fusion and improve cancer subtype prediction. Nevertheless, existing methods may still suffer from information loss during feature transformation, scalability limitations due to high computational complexity, and challenges in effectively modeling cross-modal relationships while maintaining a balance between model expressiveness and the risk of overfitting.

To address the above issues, we propose a novel algorithm called the Multi-modal Hierarchical Attention (MMHA) algorithm for disease diagnosis and subtype differentiation. Unlike previous methods, we abandon the graph-based network framework and instead use hierarchical attention to capture the complementary information between different modalities and the similarities between patients. Additionally, as diseases often follow a long-tail distribution, the model incorporates class weights for constrained optimization, thereby improving the overall performance. In summary, the main contributions of this paper are as follows:

- We propose a Multi-Modal Hierarchical Attention (MMHA) algorithm for multi-modal disease diagnosis, which effectively integrates multi-source data and is well-suited for scenarios with non-fixed modalities.
- MMHA has been validated for its effectiveness in disease diagnosis across multiple modalities, revealing that there is certain

information complementarity between the modalities, while also identifying some degree of exclusivity.

- Compared to the state-of-the-art methods, the significant improvement across six datasets from five diseases demonstrates the clear advantages of our MMHA in disease prediction tasks. Meanwhile, the visualization of the contribution scores reflected by the dependencies among multi-modalities also provides explainable decision support for doctors in real medical applications and offers insights for disease research.

2. Related works

2.1. Multi-modal information fusion

Early methods for disease prediction combined multi-modal data by directly concatenating features from different modalities (Lee et al., 2009a; Pham et al., 2007), but this approach often faced challenges such as the curse of dimensionality. Techniques like multi-kernel learning (Liu et al., 2013; Wen et al., 2017) and deep neural networks (DNNs) (Kang et al., 2020; Pan et al., 2021b) improved this process by combining kernel matrices or encoding features separately before concatenation. However, these methods often failed to capture interactions between modalities, limiting their ability to leverage complementary information.

More advanced methods, such as deep semi-nonnegative matrix factorization (SNMF) and multi-attention modules (Lee et al., 2009a; Ning et al., 2021; Zhou et al., 2019), enhanced feature fusion by preserving shared information while maintaining the integrity of the original data. Despite these advancements, these approaches often prioritized shared features over modality-specific details, which are critical for accurate diagnosis (Pan, Ding, & Chen, 2021a). Additionally, many methods overlooked patient similarities, a crucial factor in improving disease prediction accuracy.

2.2. Graph structure construction methods

Graph neural networks (GNNs) have become a prominent tool for disease diagnosis and prognosis prediction due to their ability to model relationships between data points (Kazi et al., 2019; Parisot et al., 2018, 2017; Tariq et al., 2022). Their effectiveness largely depends on graph structure construction, which can be achieved using supervised or unsupervised methods. Supervised approaches, such as MMGL (Peng et al., 2020) and HMFGL (Wang et al., 2024), compute similarities between multi-modal representations using deep learning models. MMGL employs adaptive graph learning to uncover intrinsic patient relationships and build optimized graph structures for downstream tasks. HMFGL refines graph construction by using degree-sensitive edge pruning and k-nearest neighbors (kNN) sparsification to eliminate noise, combining multi-modal embeddings with original patient features for a more accurate graph. Similarly, MOGAT (Tanvir et al., 2024) uses a multi-head attention mechanism to assign attention weights to neighboring patients, effectively capturing their influence in the graph.

Unsupervised methods, such as MoGCN (Li et al., 2022), integrate multi-omics data using autoencoders (AEs) and Shared Nearest Neighbor Fusion (SNF) (Wang et al., 2014), which merges patient similarity networks (PSNs) (Zhang et al., 2022) into a unified graph. This graph is then processed with GCNs to construct cancer subtype classification models by combining the outputs of AE and SNF. While GNNs are effective, they face significant challenges. Constructing and loading entire graph structures into memory requires substantial computational resources. Additionally, patient data from different modalities, such as imaging, genomic, and clinical data, often have distinct representations. Integrating these modalities into a single graph structure can fail to capture complex cross-modal relationships. For instance, graph edges often represent intra-modal relationships, neglecting dependencies across modalities, which limits the expressive power of the graph.

To overcome these limitations, we propose the MMHA model, a transformer-based approach that moves beyond graph-based frameworks. MMHA handles non-fixed modality inputs, effectively capturing shared information across modalities while leveraging their complementary features. This model provides a flexible and expressive solution for multi-modal disease diagnosis, addressing the challenges inherent in graph-based methods.

3. Method

In this section, we present the methodology of the proposed multi-modal hierarchical attention model (MMHA) designed for disease classification tasks. As shown in Fig. 1, MMHA comprises a single module—the hierarchical attention mechanism with a classification-weighted processing function.

3.1. Problem formulation

Let $\{X_i\}_{i=1}^N$ represent the set of feature vectors for N patients, and let $\{Y_i\}_{i=1}^N$ denote the corresponding labels. Each feature vector for patient i , denoted as $X_i = [x_i^1, x_i^2, \dots, x_i^M]$, consists of M modalities, where $x_i^m \in \mathbb{R}^{d_m}$ represents the m -th modality for patient i , and $m = 1, \dots, M$. The multi-modal inputs are processed by a hierarchical attention network that performs both feature extraction and fusion through a multi-head attention mechanism. The fused features are then passed through a series of fully connected layers, ultimately producing the final classification output, denoted as $\hat{Y}$. The subsequent subsections provide a detailed description of each module in the model.

3.2. Modality-fusion rerepresentation learning

In real diagnostic settings, relying on single-modal information can often lead to misdiagnosis. Therefore, doctors typically need to analyze multiple test results of a patient in order to make an accurate diagnosis. To enable computers to better mimic the diagnostic process of doctors, the model must be able to use multi-modal data to extract both common and complementary information across patients with different diseases. When fusing multi-modal data, it is important to capture the commonality between modalities to increase similarity within the same class,

while also preserving complementarity and appropriate differences to enhance the distinction between classes. The proposed MMHA model aims to learn a hybrid feature V from the raw multi-modal features X through the fusion function F , and we have:

$$X \xrightarrow{F_{\text{hAttn}}} \hat{V}, \quad (1)$$

where $\hat{V} \in \mathbb{R}^{N \times d_c}$ represent the multi-modality fusion feature $\hat{V}_i$ for a patient i .

To learn effective multi-modal fusion features that can capture both commonality and complementarity, the principle of the attention mechanism aligns well with the goal of multi-modal fusion. The attention mechanism has been widely proven to be effective in identifying relationships between elements at different positions in sequence or image data. Therefore, a multi-channel hierarchical attention mechanism is employed to discover inter-modal commonality and capture complementary information, thereby guiding the learning of modality-aware representations. As illustrated in Fig. 1, the raw heterogeneous multi-modal features X are processed into structured data, where each modality may have a different dimensionality. For each individual i , the multi-modal feature is represented as $x_i = \{x_i^1, x_i^2, \dots, x_i^M\}$, with the features from different modalities organized in separate hierarchical channels.

For each modality, we apply a separate linear transformation to obtain the query, key, and value. Specifically, the transformations for the queries, keys, and values are defined as:

$$Q_i^m = W_q^m x_i^m, \quad K_i^m = W_k^m x_i^m, \quad V_i^m = W_v^m x_i^m, \quad (2)$$

here, W_q^m , W_k^m , and $W_v^m \in \mathbb{R}^{d_k \times d_m}$ are the learned weight matrices, where d_m is the dimensionality of the input features for modality m , and d_k is the dimensionality of the projected query/key/value vectors used in the attention mechanism. These transformations map the features of each modality into the query, key, and value spaces, respectively, for the attention mechanism.

For the fusion of different modalities, we adopt a hierarchical approach, which is a post-fusion strategy for multi-modal integration. First, for each patient i , after the initial transformation, we obtain the corresponding query, key, and value for each modality, denoted as Q_i^m , K_i^m , and V_i^m , respectively. The next step involves calculating the

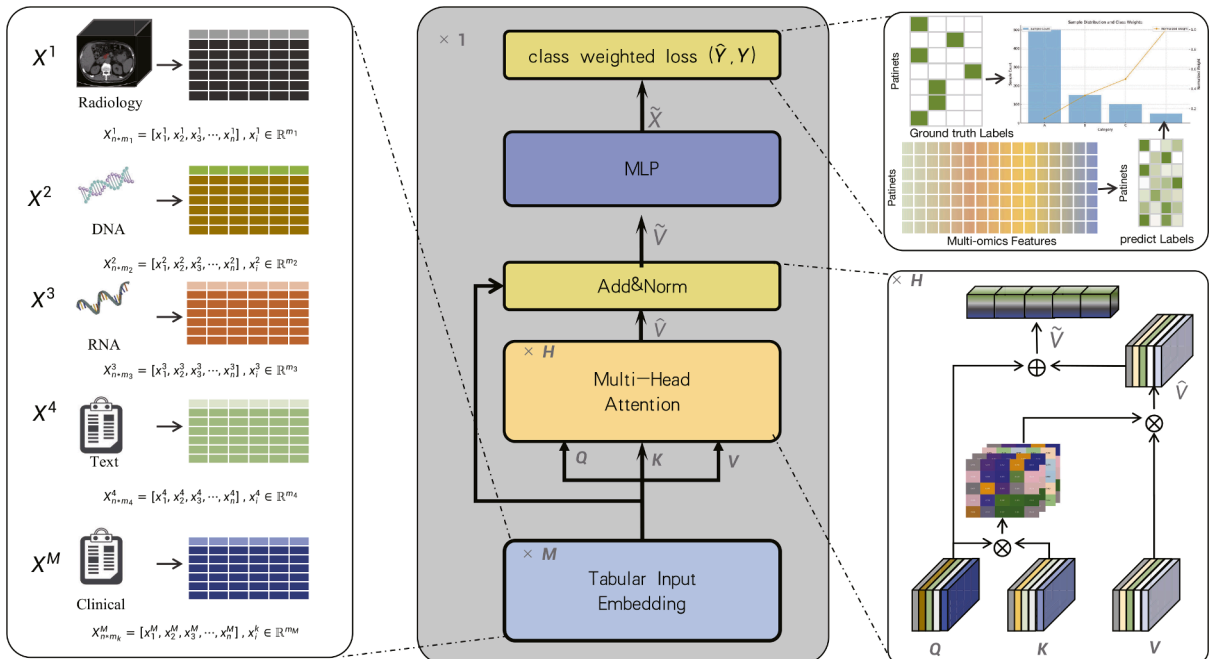


Fig. 1. The framework of MMHA. The model processes inputs from radiology, DNA, RNA, text, and clinical data to generate predictions with class-weighted loss.

intra-modality attention scores and performing feature fusion within each modality.

$$\text{Attention}(Q_i^m, K_i^m, V_i^m) = \text{softmax}\left(\frac{Q_i^m K_i^{mT}}{\sqrt{d_k}}\right) V_i^m \quad (3)$$

Using the computed attention scores, the features within each modality are updated, effectively incorporating the relevant features from that modality:

$$\hat{Q}_i^m = \text{Attention}(Q_i^m, K_i^m, V_i^m) \cdot V_i^{mT} \quad (4)$$

Subsequently, the feature representation for patient i is formed by concatenating the updated features from all modalities:

$$\hat{V}_i = \{\hat{Q}_i^1, \hat{Q}_i^2, \dots, \hat{Q}_i^M\} \quad (5)$$

To effectively integrate the features from different modalities, we compute the inter-modality attention scores, which enable the fusion of multi-modal features:

$$W_m = \text{softmax}\left(\left[\hat{Q}_i^m \cdot Q_i^{jT}\right]\right) \quad (6)$$

Here, $\hat{Q}_i^m \cdot Q_i^{jT}$ represents the pairwise dot product between different modalities. After applying the softmax function for normalization, we obtain the attention weights between the modalities, denoted as $W = \{w_1, w_2, \dots, w_M\}$. This design reflects the modality-aware property of the model in two key ways. First, although all modalities adopt the same attention framework structurally, each modality is processed through its own independent parameter set (W_q^m, W_k^m, W_v^m), which preserves the unique characteristics and statistical distributions of each modality. Second, the hierarchical structure-comprising both intra- and inter-modality attention-ensures that the model learns modality-specific representations at early stages, and selectively emphasizes complementary modality pairs at later stages. This variable-channel architecture enables the model to dynamically adapt to heterogeneity across modalities, effectively achieving modality-awareness while maintaining scalability. Based on these attention weights, the features from different modalities for the same patient are fused as follows:

$$\hat{V}_i = W V_i \quad (7)$$

Where $\hat{V}_i$ is the fused multi-modal feature representation of the patient after applying the hierarchical attention mechanism.

3.3. Class adjustment factor

In medical multi-modal learning, class imbalance is a common challenge where certain classes are underrepresented, leading to biased predictions. To address this, we introduce class adjustment weights to ensure that each class contributes appropriately during model training, regardless of its sample size.

For a patient i belonging to class c , the class adjustment weight d_i is computed as the reciprocal of the sample count for that class, normalized across all classes. Let N_c denote the number of samples in class c , and C the total number of classes. The weight d_i is defined as:

$$d_i = \frac{1/N_c}{\sum_{c=1}^C (1/N_c)} \quad (8)$$

Here, d_i ensures that underrepresented classes have higher weights, enhancing the model's focus on minority classes. Unlike approaches that consider modality-specific distributions, our class adjustment factor is solely dependent on the sample's class, making it independent of the number or type of modalities.

3.4. Optimization

In the MMHA framework, multi-modal data from different sources are fused into a unified feature representation $\hat{V}_i$ for each patient. This representation captures both shared and complementary information

across modalities. The fused multi-modal representations of all N patients, denoted as $\hat{V} \in \mathbb{R}^{N \times d}$, are passed through a multi-layer perceptron (MLP) to generate the unnormalized class scores:

$$c = \text{MLP}(\hat{V}) \quad (9)$$

Here, $c \in \mathbb{R}^{N \times C}$ is the logits matrix, where each row $c_i \in \mathbb{R}^C$ corresponds to the predicted logits for sample i . To obtain the probability distribution over classes, the logits are transformed using the LogSoftmax function:

$$c_{\log} = \text{LogSoftmax}(c, \text{dim} = 1) \quad (10)$$

For each sample i with true label $y_i \in \{1, \dots, C\}$, the per-sample loss is computed as the negative log-probability of the correct class:

$$\text{criterion}(c_{\log}, y_i) = -\log \left(\frac{\exp(c_{i, y_i})}{\sum_{j=1}^C \exp(c_{i, j})} \right) \quad (11)$$

To address class imbalance, a sample-specific weight d_i is applied, defined as $d_i = \frac{1/N_{y_i}}{\sum_{c=1}^C (1/N_c)}$. The final training loss is then calculated as the average of the weighted losses across all samples:

$$\mathcal{L}_{\text{total}} = \frac{1}{N} \sum_{i=1}^N [\text{criterion}(c_{\log}, y_i) \cdot d_i] \quad (12)$$

This optimization objective effectively balances class distribution while preserving the discriminative power of fused multi-modal features. It ensures that the MMHA model remains robust and generalizable across heterogeneous patient cohorts.

4. Experimental results and analysis

4.1. Datasets

To validate the effectiveness of our model, we applied the proposed method to five different biomedical disease subtype classification tasks using six different datasets. The dataset details are provided in Table 1. To mitigate the impact of noisy or redundant features on classification performance, preprocessing and feature selection were carried out separately for each omics data type, and the number of features used for training is also presented in Table 1.

- **TADPOLE:** TADPOLE dataset is a subset of the Alzheimer's Disease Neuroimaging Initiative (ADNI) database, which includes multi-modal features such as MRI, PET, cognitive tests, cerebrospinal fluid (CSF) biomarkers, risk factors, and demographic information. In the Alzheimer's disease prediction task, we selected 603 subjects from the TADPOLE dataset based on the data splits used in the MMGL (Peng et al., 2020) paper. These subjects were divided into 211 normal controls (NC), 320 mild cognitive impairment (MCI) patients, and 72 Alzheimer's disease (AD) patients. Within the MCI group, some patients will progress to AD over time, labeled as progressive MCI (pMCI), while others remain stable, labeled as stable MCI (sMCI).
- **ROSMAP:** The ROSMAP dataset contains multi-omics data from Alzheimer's disease (AD) patients and normal controls (NC), including mRNA expression, DNA methylation, and miRNA expression data. ROSMAP consists of two datasets, RO and MAP, both of which are longitudinal clinical-pathological cohort studies of AD from Rush University. In our experiment, we used the same data splits as those in the MOGONET (Wang et al., 2021) paper, with 169 normal controls and 182 Alzheimer's disease patients in the ROSMAP dataset.
- **ABIDE:** The Autism Brain Imaging Data Exchange (ABIDE) dataset aims to deepen the understanding of the neural basis of autism, collecting over 1000 resting-state functional magnetic resonance imaging (fMRI) data from 24 different research centers, along with corresponding phenotypic data. In the autism prediction task, we used

Table 1

Summary of datasets. The ROSMAP dataset is for the classification of Alzheimer's disease (AD) patients versus normal control (NC). The LGG dataset is for grade classification in low-grade glioma (LGG). The KIPAN dataset is for kidney cancer type classification, including chromophobe renal cell carcinoma (KICH), clear renal cell carcinoma (KIRC), and papillary renal cell carcinoma (KIRP). The BRCA dataset is for breast invasive carcinoma (BRCA) PAM50 subtype classification with normal-like, basal-like, human epidermal growth factor receptor 2 (HER2)-enriched, Luminal A, and Luminal B subtypes. The ABIDE dataset includes phenotypic and functional data for the classification of autism spectrum disorder (ASD) versus normal control (NC). The TADPOLE dataset includes imaging and cognitive biomarker data for predicting conversion from mild cognitive impairment (MCI) to Alzheimer's disease (AD).

Data Set	Type of Multi-Omics Data	Category
ABIDE	PHENO: 48, ANAT: 6, FUNC: 10, Correlation: 256	NC: 468, ASD: 403
ROSMAP	mRNA: 200, meth: 200, miRNA: 200	NC: 169, AD: 182
LGG	mRNA: 2000, meth: 2000, miRNA: 548	Grade2: 246, Grade3: 264
BRCA	mRNA: 1000, meth: 1000, miRNA: 503	Normal-like: 115, Basal-like: 131, HER2-enriched: 46, Luminal A: 436, Luminal B: 147
KIPAN	mRNA: 2000, meth: 2000, miRNA: 445	KICH: 66, KIRC: 318, KIRP: 274
TADPOLE	UCSFSX: 145, UCBERKELEYAV45: 148, UPENNBIOBK9: 3, RISK_FACTOR: 36, COGNITIVE_TEST: 24, ROI_AVERAGE: 9	NC: 211, sMCI: 275, pMCI: 45, AD: 72

the data splits from the MMGL (Peng et al., 2020) paper, selecting four types of modalities: demographic information, automated anatomical quality assessment metrics, automated functional quality assessment metrics, and fMRI connection networks. We selected 871 subjects from the ABIDE dataset, consisting of 468 normal controls (NC) and 403 autism spectrum disorder (ASD) patients.

- **LGG:** The LGG (Low-Grade Glioma) dataset comes from The Cancer Genome Atlas (TCGA) through the Broad GDAC Firehose, used for grade classification of LGG, specifically Grade 2 and Grade 3. Following the same data splits as MOGONET (Wang et al., 2021), we selected mRNA expression, DNA methylation, and miRNA expression data. There are 246 subjects in Grade 2 and 264 subjects in Grade 3.
- **BRCA:** The BRCA dataset is used for PAM50 subtype classification of invasive breast cancer. PAM50 is a 50-gene signature that classifies breast cancer into five molecular subtypes: normal-like, basal-like, HER2-enriched, Luminal A, and Luminal B. We used PAM50 subtype information from TCGA BRCA patients and followed the same data splits as in MOGONET (Wang et al., 2021), selecting mRNA expression, DNA methylation, and miRNA expression data. The subtype distribution is as follows: Normal-like: 115, Basal-like: 131, HER2-enriched: 46, Luminal A: 436, Luminal B: 147.
- **KIPAN:** The KIPAN dataset is used for kidney cancer subtype classification, including clear cell renal carcinoma (KIRC), chromophobe renal cell carcinoma (KICH), and papillary renal cell carcinoma (KIRP). Due to the distinct differences among these cancer types in the omics data, kidney cancer subtype classification is the simplest task among the classification tasks and serves as a proof-of-concept for multi-class applications. We selected mRNA expression, DNA methylation, and miRNA expression data, with KICH containing 66 patients, KIRC containing 318 patients, and KIRP containing 274 patients.

4.2. Baselines

- **MoGCN (Li et al., 2022):** The Multi-omics Graph Convolutional Network (MoGCN) model integrates multi-omics data by reducing dimensionality using autoencoders and constructing a patient similarity network through similarity network fusion methods for cancer subtype classification.

- **MOGLAM (Ouyang et al., 2023):** An interpretable multi-omics integration method that utilizes dynamic GCN with feature selection to extract omic-specific embedding information and employs an attention mechanism to achieve multi-omics integration.
- **MOINER (Zhang et al., 2024):** A multi-omics early integration method that enhances selected features via sample similarity networks, maps them into 2D image representations using feature similarity and the J-V algorithm, and utilizes an ensemble Vision Transformer for classification and biomarker discovery.
- **MoSwformer (Liu et al., 2024):** A multi-omics integration method that employs sliding window Transformer embedding for feature extraction, integrates omics-specific attention and representation learning modules to enhance fusion, and improves cancer subtype classification through effective multi-omics representation.
- **MOGONET (Wang et al., 2021):** Multi-omics Graph Convolutional Network (MOGONET) constructs a separate graph for each omics type and utilizes a Correlation Discovery Network to perform disease classification based on the label space of tensor fusion, as summarized in Table 2.
- **MMGL (Peng et al., 2020):** The Multi-Modal Graph Learning (MMGL) method applies modality-aware representation learning to explore shared and complementary information across different modalities and employs adaptive graph learning to capture latent graph structures.

4.3. Evaluation metrics and procedure

Based on previous studies (Li et al., 2022; Ouyang et al., 2023), we use commonly employed evaluation metrics to assess model performance, including Accuracy (ACC), weighted F1 score (F1_weighted), macro-averaged F1 score (F1_macro) for multi-class tasks, and AUC for binary classification tasks. Accuracy measures the proportion of correctly classified samples, F1_weighted addresses class imbalance by weighting each class's F1 score based on its support, and F1_macro computes the unweighted mean of F1 scores across all classes, treating all classes equally regardless of their frequency. AUC is a commonly used metric to evaluate classifier performance by summarizing the model's performance across all possible classification thresholds, providing a

Table 2

Summary of various deep learning models for multi-omics data analysis, highlighting their methods and code availability. The listed models include graph-based approaches and attention mechanisms tailored for tasks like disease classification and cancer subtype prediction.

Model Name	Method	Code availability
MoGCN (Li et al., 2022)	Graph convolutional networks (fixed number of channels)	https://github.com/Lifoof/MoGCN
MOGLAM (Ouyang et al., 2023)	Graph convolutional networks + Transformer (fixed number of channels)	https://github.com/Ouyang-Dong/MOGLAM
MOINER (Zhang et al., 2024)	Feature map + ViT	https://github.com/idrblab/MOINER
MoSwformer (Liu et al., 2024)	Attention	https://github.com/ShiLab-GitHub/MoSwformer
MOGONET (Wang et al., 2021)	Graph convolutional networks	https://github.com/txWang/MOGONET
MMGL (Peng et al., 2020)	Graph neural network	https://github.com/SsGood/MMGL

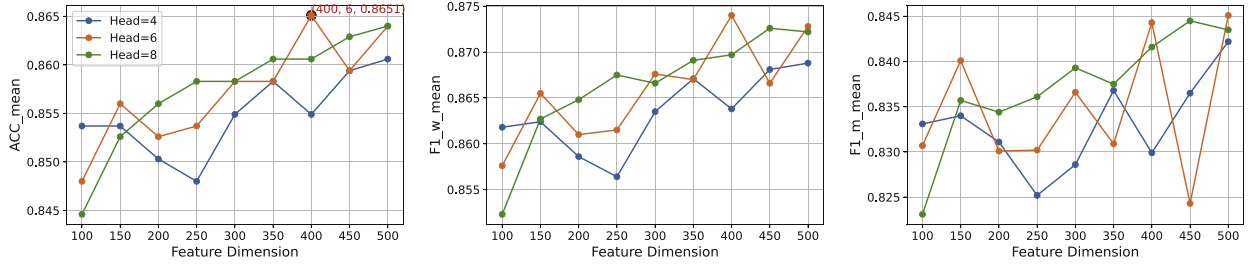


Fig. 2. The influence of different hyper-parameters on our model based on the BRCA dataset, highlighting the impact of hyper-parameters H and feature_dim on ACC, F1_w, and F1_m.

quantifiable measure of its overall discriminative ability, especially useful in binary classification tasks with imbalanced data. These metrics are calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (13)$$

$$\text{F1} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (14)$$

Here, TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. Recall is defined as $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$, and Precision is defined as $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$.

The model is evaluated using five-fold cross-validation, repeated five times to ensure robustness. Results are reported as the mean and standard deviation (mean $\pm$ std) of evaluation metrics across multiple experiments with varying random seeds. To address the imbalance in label distribution within the training data, class-specific loss weights are assigned based on the inverse of their frequencies in the training data.

4.4. Parameters analysis

In our method, there are two important hyperparameters that significantly affect the classification performance of the model. Since our

model is based on a hierarchical attention mechanism, the number of attention heads (H) and the embedding dimension of the model's features (feature_dim) are key hyperparameters that directly influence model performance. These two hyperparameters complement each other and jointly affect the model's training process. Therefore, selecting the appropriate number of heads and embedding dimension is crucial for the model's classification accuracy.

In our experiments, the hyperparameters H and feature_dim were determined through grid search on the training data. Using the BRCA dataset as an example, we demonstrated the impact of these two hyperparameters on classification performance. We tested $H = \{4, 6, 8\}$ and $\text{feature_dim} = \{100, 150, 200, 250, 300, 350, 400, 450, 500\}$. As shown in the Fig. 2, we ultimately selected $\text{feature_dim} = 400$ and $\text{head} = 6$. The hyperparameter tuning results for other experimental datasets can be found in the figures in the appendix.

4.5. Comparison experiments

As shown in Table 3, we observe that our model MMHA demonstrates outstanding performance compared to other baseline models in both binary and multiclass classification tasks. Specifically, for high-dimensional datasets (e.g., LGG and KIPAN), MMHA achieves significant performance improvements. For instance, in the LGG dataset, MMHA

Table 3

A performance comparison of various models across datasets is presented, using metrics such as Accuracy (ACC), Weighted F1 Score (F1_weight), and Area Under the Curve (AUC) for binary classification, or Macro F1 Score (F1_macro) for multi-class classification, expressed as percentages (%). The best performance for each metric is highlighted in red, while the second-best is highlighted in green.

Metrics	MoGCN (Li et al., 2022)	MOGLAM (Ouyang et al., 2023)	MOINER (Zhang et al., 2024)	MoSwformer (Liu et al., 2024)	MOGONET (Wang et al., 2021)	MMGL (Peng et al., 2020)	MMHA
ABIDE (4 modality-2 classes)							
ACC (%)	—	—	—	—	80.80 $\pm$ 2.61	88.17 $\pm$ 8.13	88.85 $\pm$ 1.80
F1_weight (%)	—	—	—	—	80.77 $\pm$ 2.51	88.06 $\pm$ 8.31	88.84 $\pm$ 1.81
AUC (%)	—	—	—	—	86.38 $\pm$ 2.65	87.99 $\pm$ 8.46	91.99 $\pm$ 0.90
ROSMAP (3 modality-2 classes)							
ACC (%)	69.81 $\pm$ 4.62	76.92 $\pm$ 4.28	76.94 $\pm$ 4.60	71.81 $\pm$ 6.64	76.08 $\pm$ 3.44	82.86 $\pm$ 4.33	85.47 $\pm$ 3.29
F1_weight (%)	69.97 $\pm$ 4.56	76.67 $\pm$ 4.53	77.04 $\pm$ 4.20	70.11 $\pm$ 10.39	76.01 $\pm$ 3.48	82.82 $\pm$ 4.35	85.44 $\pm$ 3.29
AUC (%)	69.85 $\pm$ 4.69	84.06 $\pm$ 3.34	76.99 $\pm$ 4.76	71.52 $\pm$ 7.08	79.80 $\pm$ 3.93	82.87 $\pm$ 4.54	87.41 $\pm$ 4.81
LGG (3 modality-2 classes)							
ACC (%)	72.35 $\pm$ 3.25	74.90 $\pm$ 2.37	75.69 $\pm$ 3.79	73.24 $\pm$ 4.51	75.29 $\pm$ 3.47	76.42 $\pm$ 3.66	80.88 $\pm$ 2.02
F1_weight (%)	72.70 $\pm$ 3.09	74.86 $\pm$ 2.37	74.60 $\pm$ 4.99	72.74 $\pm$ 5.76	75.16 $\pm$ 3.46	76.34 $\pm$ 3.66	80.86 $\pm$ 1.97
AUC (%)	72.07 $\pm$ 3.37	79.33 $\pm$ 3.65	75.94 $\pm$ 3.78	73.67 $\pm$ 4.22	79.91 $\pm$ 4.09	76.68 $\pm$ 3.69	83.98 $\pm$ 2.66
BRCA (3 modality-5 classes)							
ACC (%)	78.17 $\pm$ 1.32	84.23 $\pm$ 3.75	85.49 $\pm$ 2.33	83.33 $\pm$ 7.61	78.97 $\pm$ 3.26	82.72 $\pm$ 2.71	86.51 $\pm$ 2.84
F1_weight (%)	79.14 $\pm$ 1.34	85.10 $\pm$ 3.66	85.97 $\pm$ 2.40	83.03 $\pm$ 10.80	77.66 $\pm$ 3.58	81.84 $\pm$ 3.23	87.40 $\pm$ 2.63
F1_macro (%)	69.32 $\pm$ 2.71	81.68 $\pm$ 4.75	82.77 $\pm$ 2.40	77.35 $\pm$ 14.55	70.90 $\pm$ 4.90	71.89 $\pm$ 7.46	84.43 $\pm$ 4.65
KIPAN (3 modality-3 classes)							
ACC (%)	91.94 $\pm$ 1.04	97.42 $\pm$ 1.32	97.11 $\pm$ 2.44	95.07 $\pm$ 5.31	97.41 $\pm$ 1.78	97.72 $\pm$ 0.83	99.09 $\pm$ 0.89
F1_weight (%)	92.02 $\pm$ 1.03	97.42 $\pm$ 1.33	97.10 $\pm$ 2.43	94.84 $\pm$ 6.87	97.42 $\pm$ 1.78	97.73 $\pm$ 0.82	99.09 $\pm$ 0.89
F1_macro (%)	89.73 $\pm$ 1.80	96.83 $\pm$ 1.68	96.68 $\pm$ 3.31	93.48 $\pm$ 9.09	97.15 $\pm$ 1.61	96.97 $\pm$ 1.10	98.91 $\pm$ 1.21
TADPOLE (2 modality(2,5)-3 classes)							
ACC (%)	—	—	—	—	70.52 $\pm$ 4.98	90.97 $\pm$ 1.08	94.29 $\pm$ 0.98
F1_weight (%)	—	—	—	—	69.60 $\pm$ 5.00	90.95 $\pm$ 1.14	94.33 $\pm$ 0.99
F1_macro (%)	—	—	—	—	63.58 $\pm$ 5.22	88.40 $\pm$ 2.38	93.10 $\pm$ 0.96

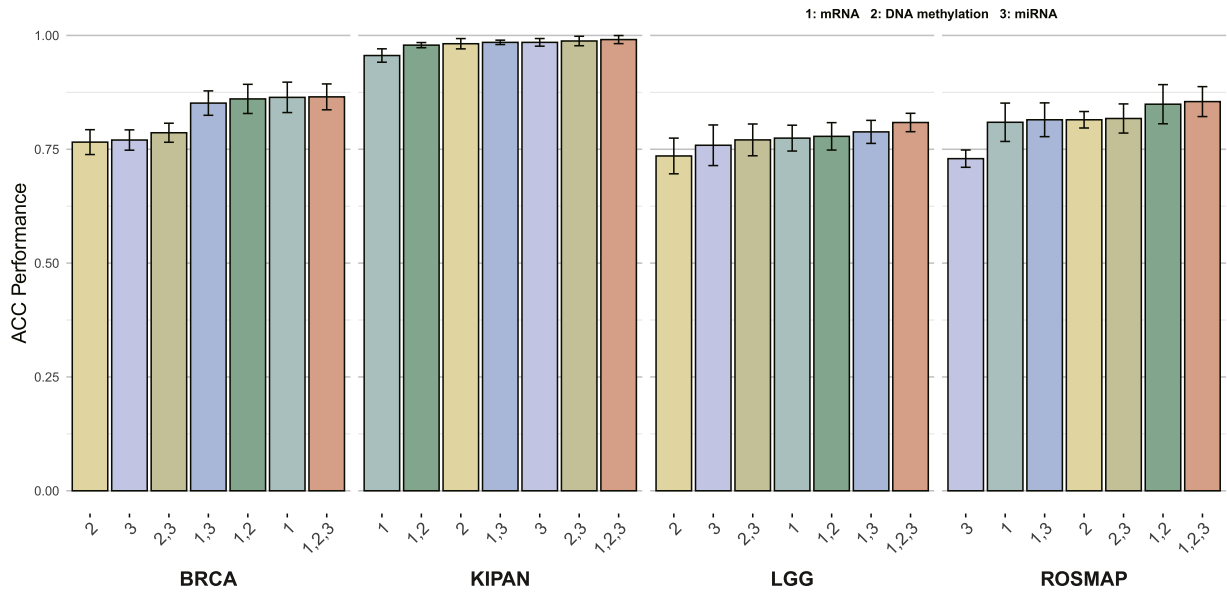


Fig. 3. Comparison of Accuracy (ACC) Performance Between Multi-Omics and Single-Omics Data. This figure compares the ACC performance across four datasets: BRCA, KIPAN, LGG, and ROSMAP. Bars represent the accuracy achieved using different single-omics data types (1: mRNA, 2: DNA methylation, 3: miRNA) and their combinations (e.g., 1 + 2, 2 + 3, 1 + 2 + 3 for multi-omics integration). Multi-omics approaches generally outperform single-omics data, with notably higher ACC performance observed in the KIPAN dataset. Error bars indicate the standard deviation of ACC values.

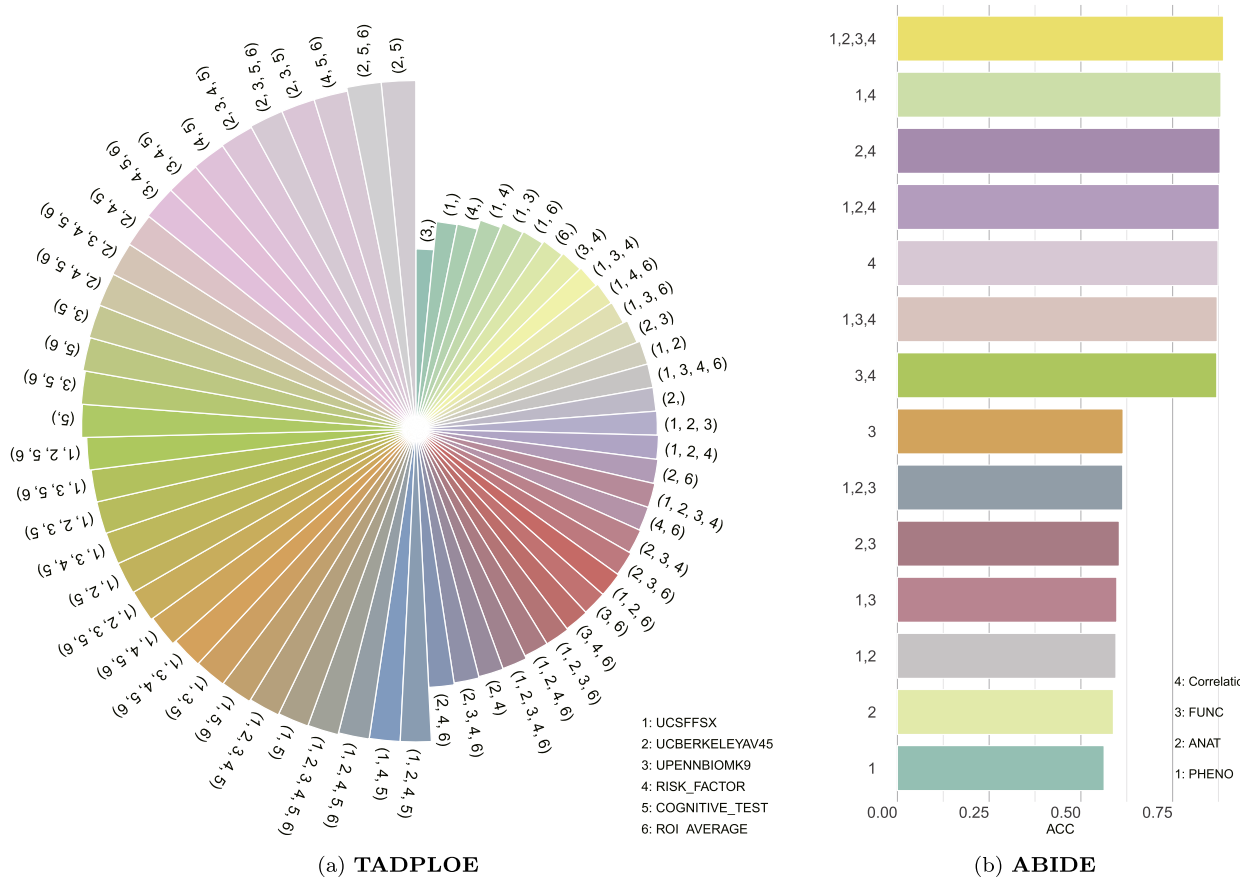


Fig. 4. Multi-omics Analysis Across TADPOLE and ABIDE Datasets. (a) TADPOLE: A radial chart illustrating the combinations of multi-omics features used for prediction, including UCSFFSX, UCBERKELEYAV45, OPENNIBMRI9, risk factors, cognitive tests, and total averages. The chart highlights various combinations, such as (1,2,3,4,5,6), with different segments reflecting their contributions. (b) ABIDE: A bar chart showing the accuracy (ACC) performance of different multi-omics feature combinations for the ABIDE dataset. Features include PHENO (1), ANAT (2), FUNC (3), and Correlation (4), with the top-performing combinations such as (1,2,3,4) and (3,4) achieving the highest accuracy.

Table 4

Comparison of model complexity (FLOPs, parameters, and computation time) between MMHA and other methods across six datasets. Here, “Others” refer to the second-best performing models in terms of predictive accuracy as reported in Table 3; for each dataset, the corresponding FLOPs and parameter counts of the second-highest scoring model (e.g., by accuracy, F1 score, or AUC) are presented.

Metrics	Method	ABIDE	ROSMAP	LGG	BRCA	KIPAN	TADPOLE
FLOPs (MFLOPs)	MMHA	0.06	6.59	59.36	64.19	10.90	6.54
	Others	0.08	39.92	195.44	8769.45	23.17	0.20
Parameters (M)	MMHA	0.05	5.30	51.33	52.64	10.18	5.53
	Others	0.05	15.54	91.34	69.85	13.78	0.07

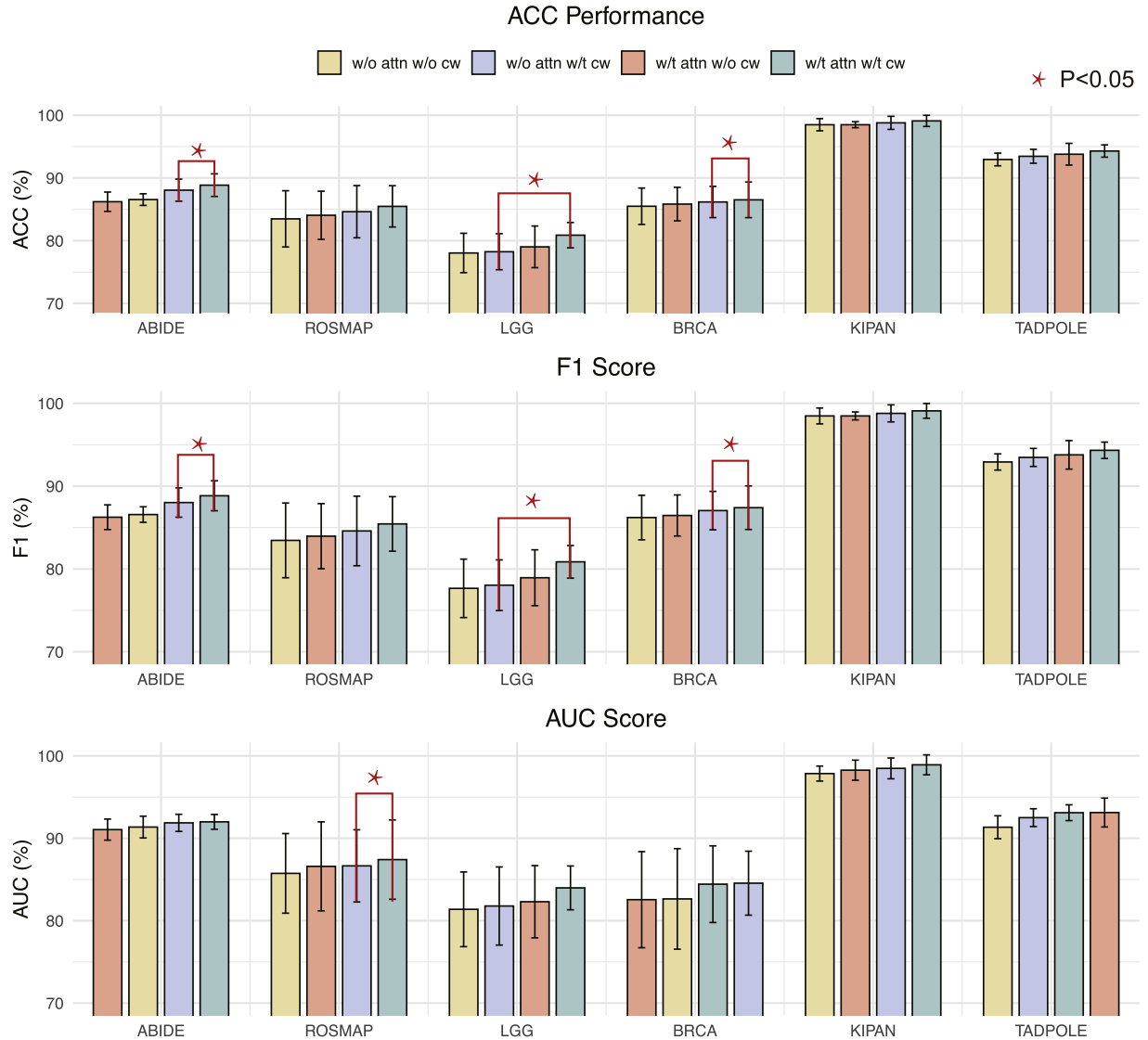


Fig. 5. Performance Metrics with Attention-based Ablation Analysis. This figure presents the performance metrics, including ACC (%), F1 (%), and AUC (%), across six datasets (ABIDE, ROSMAP, LGG, BRCA, KIPAN, and TADPOLE) under different settings. The settings involve combinations of ‘with/without attention (attn)’ and ‘with/without class weighting (cw)’. The figure emphasizes the significant performance differences between models with and without attention, with statistical significance marked for comparison.

achieves an ACC of 80.88%, improving by 4.46% compared to MMGL (Peng et al., 2020), and an F1_weight of 80.86%, improving by 4.52% over MMGL. Furthermore, MMHA achieves an AUC of 83.98%, which is 4.07% higher than MOGONET (Wang et al., 2021). In the KIPAN dataset, MMHA achieves near-perfect performance, with both ACC and F1_weight reaching 99.09%, representing improvements of 1.37% and 1.36% respectively over MMGL. Additionally, F1_macro is increased by 1.76%, rising from 97.15% with MOGONET to 98.91% with MMHA.

For low-dimensional multi-modal datasets (e.g., TADPOLE and ROSMAP), MMHA also achieves excellent results. For example, in the TADPOLE dataset, MMHA achieves an ACC of 94.29%, which is 3.32% higher than the second-best method, and an F1_weight of 94.33%, surpassing MMGL (Peng et al., 2020) by 3.38%. Additionally, its F1_macro reaches 93.10%. Similarly, in the ROSMAP dataset, MMHA achieves an ACC of 85.47%, improving by 2.61% compared to MMGL, and an F1_weight of 85.44%, showing a 2.62% improvement.

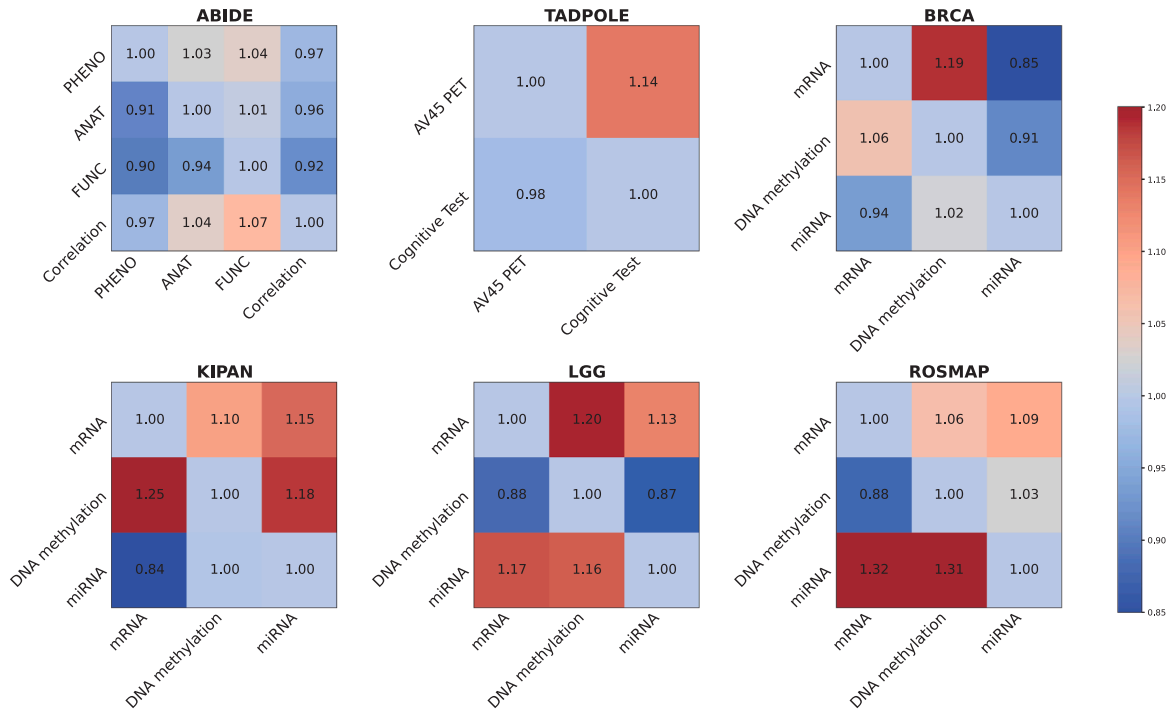


Fig. 6. Heatmaps of attention matrices across six datasets. This figure illustrates the cross-modal attention weights among different modalities for each dataset, where color intensity represents the strength of inter-modality information flow. The attention matrices are symmetrized to ensure stability. The color scale ranges from 0.85 to 1.2, where values greater than 1.0 indicate stronger cross-modal complementarity. These results demonstrate distinct patterns of multi-modal interaction aligned with disease-specific characteristics.

Furthermore, MMHA achieves an AUC of 87.41 %, which is 3.35 % higher than MOGLAM.

When applied to the BRCA dataset (a five-class classification task), MMHA demonstrates significant advantages in handling complex tasks. Specifically, MMHA achieves an ACC of 86.51 %, which is 1.02 % higher than the second-best method MOINER, an F1_weight of 87.40 %, improving by 1.43 %, and an F1_macro of 84.43 %, which is 1.66 % higher than MOINER. For the ABIDE dataset (a binary classification task), MMHA also outperforms MMGL and MOGONET. Specifically, MMHA achieves an ACC of 88.85 %, improving by 0.68 % compared to MMGL, and an F1_weight of 88.84 %, also 0.78 % higher than MMGL. Furthermore, MMHA achieves an AUC of 91.99 %, which is 4.00 % higher than MMGL.

Overall, MMHA demonstrates excellent performance in both binary and multiclass classification tasks. Whether applied to high-dimensional datasets (e.g., LGG and KIPAN) or low-dimensional multi-modal datasets (e.g., TADPOLE and ROSMAP), MMHA significantly outperforms baseline methods, fully showcasing its advantages in multi-modal feature integration and classification tasks. The adaptive weighting mechanism of MMHA enables it to fully leverage information from different omics modalities, significantly enhancing classification accuracy. Additionally, MMHA demonstrates strong robustness even in scenarios with imbalanced data distribution or small sample sizes. Furthermore, in high-dimensional tasks with a large number of features, MMHA effectively extracts critical information to improve model performance.

To quantitatively evaluate the computational efficiency and deployment feasibility of MMHA, we conducted a comparative analysis of model complexity-measured in terms of floating point operations (FLOPs) and parameter counts-across six publicly available biomedical datasets, as summarized in Table 4. MMHA consistently demonstrates a lightweight architecture, achieving substantial reductions in FLOPs compared to state-of-the-art baselines. Notably, on the BRCA dataset, MMHA requires only 64.19 MFLOPs, in stark contrast to the 8769.45 MFLOPs consumed by existing approaches. Similar trends are observed in other datasets such as ROSMAP (6.59 vs. 39.92 MFLOPs)

and LGG (59.36 vs. 195.44 MFLOPs). Furthermore, MMHA maintains a compact parameter footprint across all evaluated datasets-for example, using only 51.33M parameters on LGG compared to 91.34M in baseline models-without compromising predictive performance.

The substantial reduction in computational overhead achieved by MMHA highlights its superior efficiency and underscores its suitability for deployment in resource-constrained or real-time clinical settings. The favorable balance between computational cost and predictive accuracy positions MMHA as a compelling candidate for practical, scalable, and sustainable applications in multi-modal biomedical data analysis.

4.6. Performance of MMHA under different omics data type

To demonstrate the effectiveness of multiomics integration in improving the performance of disease subtype diagnosis tasks, we conducted a comparative analysis evaluating the MMHA model across all modality combinations in six datasets, as so in Fig. 3. For the BRCA, KIPAN, LGG, and ROSMAP datasets, which are based on mRNA, meth, and miRNA modalities, and for the ABIDE dataset using four modalities, MMHA consistently achieved the best performance in both multi-class and binary classification tasks.

Through the combination of modality inputs, we observed that different modalities contribute differently to the diagnosis of disease subtypes. For the BRCA and LGG datasets, mRNA as a single modality demonstrated greater effectiveness in subtype diagnosis. In contrast, for KIPAN and ROSMAP datasets, miRNA showed the best performance in diagnosing kidney cancer subtypes, while DNA proved most effective in determining whether the subject had AD.

Interestingly, for the TADPOLE dataset, as shown in Fig. 4, which focuses on identifying different stages of disease progression, the MMHA model revealed that the combination of UCBERKELEYAV45 features from PET imaging and modality features representing cognitive tests was the most effective in assessing AD progression.

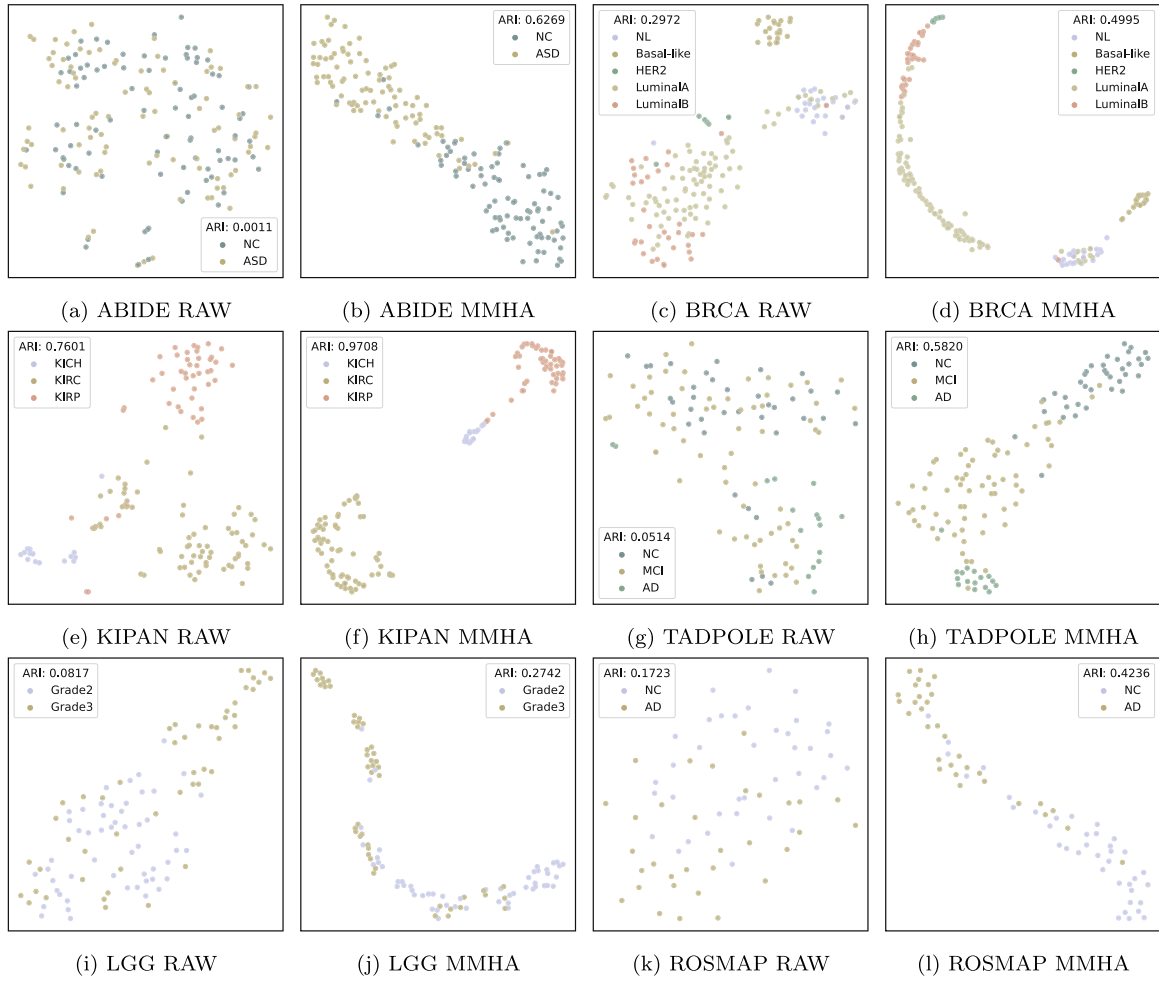


Fig. 7. Comparison of Raw and MMHA Processed Data Across Multiple Datasets. Scatter plots illustrate clustering results for six datasets (ABIDE, BRCA, KIPAN, TADPOLE, LGG, and ROSMAP) in their raw form (left column) and after MMHA processing (right column). Adjusted Rand Index (ARI) values are displayed for each plot to quantify clustering quality.

4.7. Ablation experiments

To validate the effectiveness of the hierarchical attention mechanism and class balancing, we replaced the hierarchical attention mechanism with MLP and tested the impact of whether the class distribution of the data was corrected for bias. As shown in Fig. 5, across all datasets, the model without the hierarchical attention mechanism and class balancing performed the worst, while the model utilizing both the hierarchical attention mechanism and class balancing achieved the best performance, with the improvement being most prominent in the ABIDE dataset. The hierarchical attention mechanism is particularly sensitive to whether the class distribution is balanced. When the dataset is not balanced, the hierarchical attention mechanism has little to no positive impact on model performance and can even degrade it, as observed in the ABIDE model. However, class balancing consistently improves the model's performance.

To assess the impact of the attention mechanism, we conducted independent t-tests comparing models trained with (w/t attn w/t cw) and without attention (w/o attn w/t cw) across six datasets (ABIDE, ROSMAP, LGG, BRCA, KIPAN, and TADPOLE). Statistically significant improvements were observed in ABIDE, ROSMAP, LGG and BRCA, suggesting that attention mechanisms substantially enhanced model performance in datasets characterized by complex and heterogeneous multi-modal information. In contrast, no significant differences were detected for KIPAN and TADPOLE ($p > .1$ for all metrics), likely due to their already high baseline accuracies ($> 90\%$), which limited the

detectability of further improvements. Nevertheless, even minor performance gains (e.g., $0.5\% - 2\%$) can be practically meaningful in clinical and biomedical settings, particularly when dealing with highly imbalanced or heterogeneous data.

4.8. Cross-modal attention analysis

Beyond enhancing predictive accuracy, the attention mechanism served as a critical component for modeling latent interactions across heterogeneous data modalities. By analyzing the averaged cross-modal attention matrices obtained from 5-fold cross-validation, we observed disease-specific patterns of modality synergy across six datasets, as shown in Fig. 6. For instance, in both ABIDE and TADPOLE, strong attention flows were detected between imaging-derived features and cognitive or functional modalities. Specifically, functional MRI (FUNC) in ABIDE attended prominently to both structural MRI (ANAT; 1.04) and functional connectivity (Correlation; 1.07), reflecting structural-functional coupling relevant to autism. In TADPOLE, AV45 PET strongly focused on cognitive assessments (1.14), with a slightly lower reciprocal response (0.98), suggesting an asymmetric but meaningful complementarity in modeling Alzheimer's disease progression. In the molecular datasets, attention trends reflected known regulatory hierarchies. In both BRCA and KIPAN, DNA methylation exhibited the highest attention toward mRNA (1.19 and 1.25, respectively), supporting the dominant role of epigenetic regulation in tumor subtype determination. Notably, miRNA showed limited interaction in

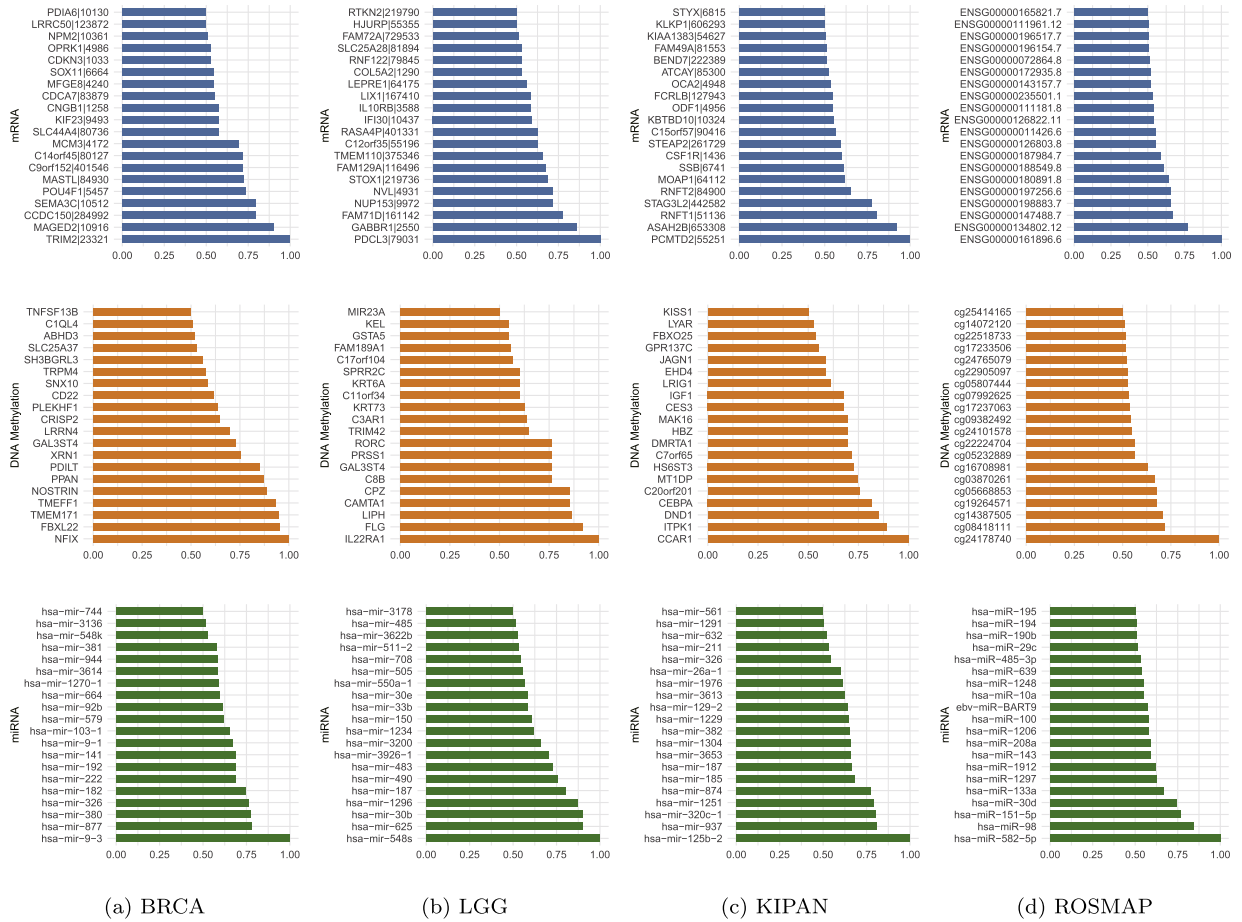


Fig. 8. Top Contributing Features Across Different Omics Modalities for BRCA, LGG, KIPAN, and ROSMAP. This figure displays the top-ranked features across three omics modalities: mRNA (blue, top row), DNA methylation (orange, middle row), and miRNA (green, bottom row) for four datasets: (a) BRCA, (b) LGG, (c) KIPAN, and (d) ROSMAP. The horizontal bars represent feature importance or contribution scores, with higher scores indicating greater relevance to the classification task. The identified features are labeled with corresponding gene names or identifiers, highlighting key biomarkers within each dataset and modality.

BRCA (0.85), but substantial engagement in KIPAN (1.18), indicating dataset-specific regulatory contributions. Similarly, in LGG, miRNA displayed strong synergy with both mRNA (1.17) and DNA methylation (1.16), emphasizing post-transcriptional regulation as a key driver in glioma stratification. ROSMAP demonstrated the overall strongest cross-modal interactions, with miRNA attending to both mRNA and methylation features (attention > 1.31), aligning with its well-established role in cognitive impairment and neurodegeneration. These findings collectively underscore the context-dependent nature of cross-modal dependencies and highlight the utility of attention mechanisms in revealing biologically meaningful relationships across diverse data types.

It is important to note, however, that high cross-modal attention values do not necessarily guarantee improved classification performance. While elevated attention weights suggest strong information exchange and potential complementarity, predictive outcomes also depend on the intrinsic discriminative strength of each modality. In cases where high-attention modalities contribute weak independent signals, performance gains may be minimal. Therefore, both inter-modality synergy and per-modality informativeness should be jointly considered in the design and interpretation of multimodal fusion frameworks.

4.9. Visualization of the modality-specified representation

An effective representation learning method should effectively preserve the original patient information in the feature space. To visually evaluate the capability of the multi-modal fusion representation learned by the MMHA model, t-SNE was applied to visualize it in two-

dimensional space for all datasets. As shown in the Fig. 7, the representations learned by the MMHA model can be clustered into several groups corresponding to different categories across the six experimental datasets. Moreover, MMHA achieved higher ARI scores [45], indicating its strong performance in multimodal data analysis.

For the TADPOLE dataset, the MMHA model selected two modalities as input and extracted multi-modal fused representations constrained by three categories. The results show that the order among clusters corresponding to different disease stages was well preserved. Additionally, for datasets such as KIPAN with higher classification accuracy, the representations learned by MMHA exhibited stronger discriminative power, characterized by smaller within-class scatter and larger between-class scatter. This demonstrates that the multi-modal representations learned by MMHA can accurately capture intra-class similarities while reflecting inter-class differences.

4.10. Investigation the interpretability of MMHA

Based on the hierarchical attention mechanism introduced in the methods section, we obtained the ranking of important biomarkers identified by MMHA from the weights of the Q layer. The top 20 important genes for the BRCA, LGG, KIPAN, and ROSMAP datasets are reported in Fig. 8. Due to the lower feature dimensions in the TADPOLE and ABIDE datasets, these datasets require modality selection for multi-modal fusion rather than feature selection when necessary. This part has already been reported in the Ablation section. For the BRCA dataset, SEMA3C, MASTL, and MCM3 from the mRNA modality, KIF23 from the DNA methylation modality, and CDKN3 from the miRNA modality have been

widely reported in studies on breast cancer mechanisms (Tian, Lin, & Yu, 2017). In the classification and grading dataset for low-grade glioma (LGG), COL5A2 and GABBR1 are derived from the mRNA modality, C3AR1 and LEPRE1 from the DNA methylation modality, and MIR23A from the miRNA modality, all of which have been widely reported in studies on disease mechanisms. Among these, MIR23A is considered the most important biomarker due to its high reporting frequency in the miRNA modality and its well-documented role in regulating tumor cell proliferation, differentiation, and drug resistance, highlighting its significant biological and clinical relevance. In the KIPAN dataset, PCMTD2 and CSF1R are derived from the mRNA modality, CCAR1 from the DNA methylation modality, and hsa-mir-125b-2 and hsa-mir-26a-1 from the miRNA modality. According to the Human Protein Atlas (HPA) (Colwill, Group, & Gräslund, 2011) database, PCMTD2 has been identified as a prognostic marker in Kidney Renal Clear Cell Carcinoma and Ovary Serous Cystadenocarcinoma. For the ROSMAP dataset, we used the publicly available MOGONET (Wang et al., 2021) dataset. Since ID-to-name mapping was not performed, this study does not include functional analysis validation.

5. Conclusion

In this paper, we propose a multi-modal graph learning framework for disease subtype prediction, named MMHA. To capture the shared and complementary information across modalities, we introduce a hierarchical attention-based multi-modal fusion model. This model aims to obtain both modality-specific and modality-shared representations while considering inter-modal correlations. Unlike previous approaches, we completely discard the graph structure and instead develop a lightweight representation learning method. Our experiments demonstrate that the simple attention mechanism effectively captures the intrinsic relationships between samples and identifies key therapeutic or diagnostic biomarkers for different cancers. Additionally, we validate that the multi-modal representations learned by MMHA provide a novel feature representation that can distinguish different subtypes. MMHA is optimized end-to-end and achieves state-of-the-art (SOTA) results across six publicly available datasets for five diseases, as verified in our experiments. In addition to its predictive accuracy, MMHA features a lightweight architecture with significantly reduced FLOPs and parameter counts, making it well-suited for deployment in resource-limited or real-time clinical settings. Nonetheless, the current model assumes complete modality availability and lacks interpretability rooted in biological priors. Future work will focus on enhancing robustness to missing modalities and incorporating domain knowledge to improve interpretability and clinical relevance. We also continue to explore MMHA's translational potential, particularly in biomarker discovery and drug target identification in prospective clinical studies.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Xue Wang reports financial support was provided by Sponsored by Shanghai Pujiang Program. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Xue Wang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration, Funding acquisition.

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References

- Arya, N., & Saha, S. (2022). Multi-modal classification for human breast cancer prognosis prediction: Proposal of deep-learning based stacked ensemble model. *IEEE/ACM Transactions on Computational Biology and Bioinformatics / IEEE, ACM*, 19, 1032–1041. <https://doi.org/10.1109/TCBB.2020.3018467>
- Barnes, D. C., Murcia, F. J. M., Mesa, C. J., González, D. S., De Inestrosa, J. R. P., Sáez, J. M. G. et al. (2023). Nonlinear weighting ensemble learning model to diagnose parkinson's disease using multimodal data. *International Journal of Neural Systems*, 33(8). <https://doi.org/10.1142/S0129065723500417>
- Boehm, K. M., Khosravi, P., Vanguri, R., Gao, J., & Shah, S. P. (2022). Harnessing multi-modal data integration to advance precision oncology. *Nature Reviews Cancer*, 22(2), 114–126. <https://doi.org/10.1038/s41568-021-00408-3>
- Cao, M. et al. (2021). Using deepGCN to identify the autism spectrum disorder from multi-site resting-state data. *Biomedical Signal Processing and Control*, 70. (Art. no. 103015).
- Chen, M., Wei, Z., Huang, Z., Ding, B., & Li, Y. (2020). Simple and deep graph convolutional networks. *Proceedings of the International Conference on Machine Learning*, (pp. 1725–1735). <https://proceedings.mlr.press/v119/chen20v.html>
- Colwill, K., Group, R. P. B. W., & Gräslund, S. (2011). A roadmap to generate renewable protein binders to the human proteome. *Nature Methods*, 8(7), 551–558. <https://doi.org/10.1038/nmeth.1607>
- Craddock, C. et al. (2013). The neuro bureau preprocessing initiative: Open sharing of preprocessed neuroimaging data and derivatives. *Frontiers Neuroinformatics*, 7.
- Du, X., & Zhao, Y. (2023). Multimodal adversarial representation learning for breast cancer prognosis prediction. *Computers in Biology and Medicine*, 157, 106765. <https://doi.org/10.1016/j.combiomed.2023.106765>
- Gao, J. et al. (2020). Mgnn: A multimodal graph neural network for predicting the survival of cancer patients. In *SIGIR '20: Proceedings of the 43rd International ACM SIGIR conference on research and development in information retrieval* (pp. 1697–1700).
- Kang, H. et al. (2020). Diagnosis of coronavirus disease 2019 (COVID-19) with structured latent multi-view representation learning. *IEEE Transactions on Medical Imaging*, 39(8), 2606–2614.
- Kazi, A. et al. (2019). Graph convolution based attention model for personalized disease prediction. In *Medical image computing and computer assisted intervention-MICCAI 2019* (pp. 122–130).
- Lan, W., Liao, H., Chen, Q., Zhu, L., Pan, Y., & Chen, Y. P. P. (2024). Deepkegg: A multi-omics data integration framework with biological insights for cancer recurrence prediction and biomarker discovery. *Briefings in Bioinformatics*, 25(3), bbae185.
- Lan, W., Yang, T., Chen, Q., Zhang, S., Dong, Y., Zhou, H., & Pan, Y. (2023). Multiview subspace clustering via low-rank symmetric affinity graph. *IEEE Transactions on Neural Networks and Learning Systems*, 35(8).
- Lee, G. et al. (2009a). A knowledge representation framework for integration, classification of multi-scale imaging and non-imaging data: Preliminary results in predicting prostate cancer recurrence by fusing mass spectrometry and histology. *2009 IEEE International symposium on biomedical imaging: From nano to macro*, (pp. 77–80).
- Li, X., Ma, J., Leng, L., Han, M., Li, M., He, F., & Zhu, Y. (2022). MoGCN: A multi-omics integration method based on graph convolutional network for cancer subtype analysis. *Frontiers in Genetics*, 13, 806842. CrossRef, PubMed. <https://doi.org/10.3389/fgene.2022.806842>
- Liu, F., Zhou, L., Shen, C., & Yin, J. (2013). Multiple kernel learning in the primal for multimodal alzheimer's disease classification. *IEEE Journal of Biomedical and Health Informatics*, 18(3), 984–990. <https://doi.org/10.1109/JBHI.2013.2285378>
- Liu, M., Zhu, Y., Li, X., Shi, X., & Wang, L. (2024). Moswformer: A transformer-based patient classification model using multiomics data. In *Fourth International conference on biomedicine and bioinformatics engineering (ICBBE 2024)* (pp. 276–280). SPIE (vol. 13252).
- Marinescu, R. V. et al. (2018). Tadpole challenge: Prediction of longitudinal evolution in alzheimer's disease. In *Predictive Intelligence in Medicine: Second International Workshop, PRIME 2019, Held in Conjunction with MICCAI 2019* (pp. 1–10). Shenzhen, China: Springer.
- Ning, Z. et al. (2021). Relation-induced multi-modal shared representation learning for alzheimer's disease diagnosis. *IEEE Transactions on Medical Imaging*, 40(6), 1632–1645. <https://doi.org/10.1109/TMI.2021.3066907>
- Osipov, A., Nikolic, O., Gertych, A., Parker, S., Hendifar, A., Singh, P., Filippova, D., Dagliyan, G., Ferrone, C. R., Zheng, L. et al. (2024). The molecular twin artificial-intelligence platform integrates multi-omic data to predict outcomes for pancreatic adenocarcinoma patients. *Nature Cancer*, 5(2), 299–314.
- Ouyang, D., Liang, Y., Li, L., Ai, N., Lu, S., Yu, M., Liu, X., & Xie, S. (2023). Integration of multi-omics data using adaptive graph learning and attention mechanism for patient classification and biomarker identification. *Computers in Biology and Medicine*, 164, 107303. <https://doi.org/10.1016/j.combiomed.2023.107303>
- Pan, Q., Ding, K., & Chen, D. (2021a). Multi-classification prediction of alzheimer's disease based on fusing multi-modal features, *2021 IEEE International Conference on Data Mining (ICDM)*. IEEE Auckland, New Zealand, (pp. 1270–1275).

- Pan, X. et al. (2021b). Multi-view separable pyramid network for AD prediction at MCI stage by 18f-FDG brain PET imaging. *IEEE Transactions on Medical Imaging*, 40(1), 81–92.
- Parisot, S., Ktena, S. I., Ferrante, E., Lee, M., Guerrero, R., Glocker, B., & Rueckert, D. (2018). Disease prediction using graph convolutional networks: Application to autism spectrum disorder and alzheimer's disease. *Medical Image Analysis*, 48, 117–130. <https://doi.org/10.1016/j.media.2018.06.001>
- Parisot, S., Ktena, S. I., Ferrante, E., Lee, M., Moreno, R. G., Glocker, B., & Rueckert, D. (2017). Spectral graph convolutions for population-based disease prediction. In *Medical image computing and computer assisted intervention-MICCAI 2017* (pp. 177–185).
- Peng, P., Zhang, Y., Ju, Y., Wang, K., Li, G., Calhoun, V. D., & Wang, Y. P. (2020). Group sparse joint non-negative matrix factorization on orthogonal subspace for multimodal imaging genetics data analysis. *IEEE/ACM Transactions on Computational Biology and Bioinformatics / IEEE, ACM*, 19(1), 479–490. <https://doi.org/10.1109/TCBB.2020.2999397>
- Pham, T.-T. et al. (2007). Latent semantic fusion model for image retrieval and annotation. In *Proceedings of the 16th ACM conference on Conference on information and knowledge management, CIKM*, pp. 439–444).
- Qian, X., Pei, J., Han, C., Liang, Z., Zhang, G., Chen, N., Zheng, W., Meng, F., Yu, D., Chen, Y. et al. (2025). A multimodal machine learning model for the stratification of breast cancer risk. *Nature Biomedical Engineering*, 9(3), 356–370.
- Tanvir, R. B. et al. (2024). Mogat: A multi-omics integration framework using graph attention networks for cancer subtype prediction. *International Journal of Molecular Sciences*, 25(5), 2788. <https://doi.org/10.3390/ijms25052788>
- Tariq, A. et al. (2022). Graph-based fusion modeling and explanation for disease trajectory prediction. *medRxiv*, 2022–10.
- Tian, J., Lin, Y., & Yu, J. (2017). E2f8 Confers cisplatin resistance to ER+ breast cancer cells via transcriptionally activating MASTL. *Biomedicine & Pharmacotherapy*, 92, 919–926.
- Vanguri, R. S., Luo, J., Aukerman, A. T., Egger, J. V., Fong, C. J., Horvat, N., Pagano, A., de Arimateia Batista, A.-F. J., Geneslaw, L., Rizvi, H. et al. (2022). Multimodal integration of radiology, pathology and genomics for prediction of response to PD-(L) 1 blockade in patients with non-small cell lung cancer. *Nature cancer*, 3(10), 1151–1164.
- Wang, B. et al. (2014). Similarity network fusion for aggregating data types on a genomic scale. *Nature Methods*, 11, 333–337. <https://doi.org/10.1038/nmeth.2810>
- Wang, R. et al. (2024). Hybrid multimodal fusion for graph learning in disease prediction. *Methods*, 229, 41–48. Epub 2024 Jun 14, PMID: 38880433. <https://doi.org/10.1016/j.jymeth.2024.06.003>
- Wang, T., Shao, W., Huang, Z., Tang, H., Zhang, J., Ding, Z., & Huang, K. (2021). Mogonet integrates multi-omics data using graph convolutional networks allowing patient classification and biomarker identification. *Nature Communications*, 12, 3445. CrossRef. <https://doi.org/10.1038/s41467-021-23768-2>
- Wen, H., Liu, Y., Rekik, I., Wang, S., Chen, Z., Zhang, J., Zhang, Y., Peng, Y., & He, H. (2017). Multimodal multiple kernel learning for accurate identification of tourette syndrome children. *Pattern Recognition*, 63, 601–611. <https://doi.org/10.1016/j.patcog.2016.09.039>
- Yin, C., Cao, Y., Sun, P., Zhang, H., Li, Z., Xu, Y., & Sun, H. (2022). Molecular subtyping of cancer based on robust graph neural network and multi-omics data integration. *Frontiers in Genetics*, 13, 884028. CrossRef, PubMed. <https://doi.org/10.3389/fgene.2022.884028>
- Ying, Y., Yang, T., & Zhou, H. (2023). Multimodal fusion for alzheimer's disease recognition. *Applied Intelligence*, 53(12), 16029–16040. <https://doi.org/10.1007/s10489-022-04255-z>
- Zhang, D., Wang, Y., Zhou, L., Yuan, H., Shen, D., Initiative, A. D. N. et al. (2011). Multimodal classification of alzheimer's disease and mild cognitive impairment. *NeuroImage*, 55(3), 856–867. <https://doi.org/10.1016/j.neuroimage.2011.01.008>
- Zhang, G. et al. (2022). A novel liver cancer diagnosis method based on patient similarity network and denseGCN. *Scientific Reports*, 12, 6797. <https://doi.org/10.1038/s41598-022-10441-3>
- Zhang, W., Mou, M., Hu, W., Lu, M., Zhang, H., Zhang, H., Luo, Y., Xu, H., Tao, L., Dai, H. et al. (2024). Moirer: A novel multiomics early integration framework for biomedical classification and biomarker discovery. *Journal of Chemical Information and Modeling*, 64(7), 2720–2732.
- Zheng, X., Tian, S., Li, T., Zhang, S., Zhou, X., Liu, Y., Su, R., Zhang, M., Li, B., Qi, C. et al. (2025). Host FSTL1 defines the impact of stem cell therapy on liver fibrosis by potentiating the early recruitment of inflammatory macrophages. *Signal Transduction and Targeted Therapy*, 10(1), 81.
- Zhou, H.-Y., Yu, Y., Wang, C., Zhang, S., Gao, Y., Pan, J., Shao, J., Lu, G., Zhang, K., & Li, W. (2023). A transformer-based representation-learning model with unified processing of multimodal input for clinical diagnostics. *Nature Biomedical Engineering*, (pp. 1–13). <https://doi.org/10.1038/s41551-023-01045-x>
- Zhou, T. et al. (2019). Deep multi-modal latent representation learning for automated dementia diagnosis. In *Medical image computing and computer-assisted intervention. (MICCAI 2019)* (pp. 629–638). Cham, Switzerland: Springer.
- Zhu, Q., Du, B., & Yan, P. (2019). Multi-hop convolutions on weighted graphs. *arXiv preprint arXiv:1911.04978*.